

BANGLADESH UNIVERSITY OF PROFESSIONALS

FACULTY OF SCIENCE AND TECHNOLOGY

DEPT. OF COMPUTER SCIENCE AND ENGINEERING (CSE)

Project Report

Title: Arduino-Based Rain Detection and Automatic Cloth Protection System.

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1. Introduction

The rapid growth of automation and smart home systems has enabled the development of solutions that improve convenience, safety, and efficiency in everyday life. One such application is the automation of outdoor clothes-drying systems, where weather conditions such as sudden rain, low sunlight, or inconsistent drying time can cause inconvenience and potential damage to clothes. Traditional manual systems require users to constantly monitor weather changes and adjust the position of the clothes manually, making the process time-consuming and unreliable.

To address these challenges, this project presents an Automated Clothes Drying System that intelligently monitors environmental conditions and controls the movement of clothes using a motorized mechanism. The system utilizes an LDR sensor to measure sunlight intensity, a rain sensor to detect precipitation, and a DS3231 Real-Time Clock (RTC) module to accurately track drying time. An Arduino microcontroller processes the sensor data and applies decision-making logic to determine whether the clothes should remain outside for drying or be brought under shelter.

A 28BYJ-48 stepper motor, driven by a ULN2003 motor driver, is used to physically move the clothes rail between the outdoor and sheltered positions. The integration of sensors, timing logic, and automated motor control ensures that the clothes receive optimal sunlight exposure while being protected from rain or low-light conditions. This system provides a low-cost, reliable, and energy-efficient solution to an everyday household problem, making it suitable for both urban and rural environments.

Overall, this project demonstrates how simple sensors and embedded systems can be combined to create an effective automated solution that improves convenience and reduces user intervention in the drying process.

2. Methodology

This project implements an automated clothes-drying system that can detect sunlight, rainfall, and total drying time, and then control a stepper motor to move clothes between an outdoor drying position and an indoor sheltered position. The methodology involves four main components: sensor data acquisition, real-time time tracking using an RTC module, decision-making logic, and motor actuation.

2.1 System Initialization

At system startup, the following steps are performed:

2.1.1 Stepper Motor Setup

- A 28BYJ-48 stepper motor with ULN2003 driver is initialized using the Stepper library.
- Motor speed is set to 10 RPM.
- The motor is configured to rotate in full revolutions, where one revolution consists of 2048 steps.

2.1.2 RTC Module Initialization

- A DS3231 Real-Time Clock (RTC) module is initialized using the RTClib library.
- If the RTC has lost power or been reset, the module automatically adjusts to the system's compile-time to restore accurate timing.
- The system records the starting timestamp (startTimeEpoch) to track how long the clothes have remained in sunlight.

2.1.3 Initial Drying-State Setup

- The variable *isSunDrying* is set to 1 (true), assuming the clothes begin in the drying position outdoors.
- Thresholds for the LDR (light sensor) and rain sensor are defined to determine weather conditions.

2.2 Sensor Data Acquisition

The system continuously reads environmental data through the Arduino's analog pins:

2.2.1 LDR Sensor (Sunlight Detection)

- LDR is connected to analog pin A2.
- Low readings indicate reduced sunlight (shade or cloudy conditions).
- The reading is compared against a predefined threshold (LDRthreshold).

2.2.2 Rain Sensor (Rain Detection)

- Rain sensor is connected to analog pin A0.
- Lower values indicate presence of rain.
- The reading is compared to rainSensorthreshold.

These readings allow the system to dynamically assess environmental conditions.

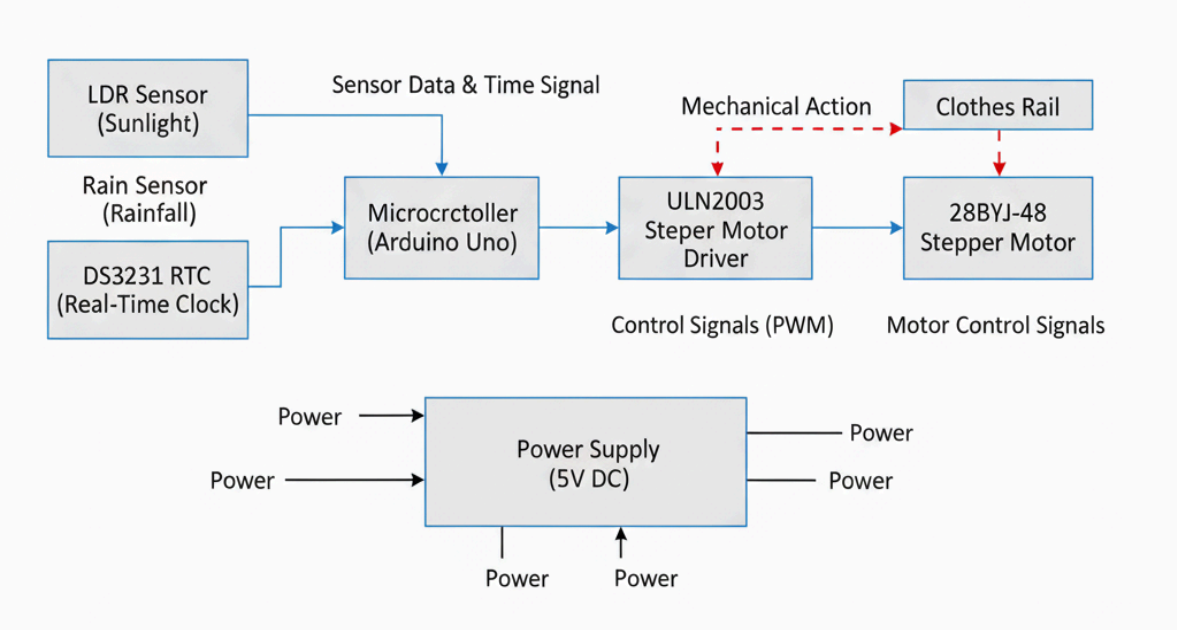


Fig: System Architecture Diagram

2.3 Time Tracking Using RTC

The drying time is calculated using the DS3231 RTC:

- **startTheClock()**
  Records the epoch time when the clothes are placed in sunlight.
- **stopTheClock()**
  Updates total sun-exposure time (staingAtSun) when the clothes are moved to shelter.
- **isDried()**
  Checks if total accumulated drying time (staingAtSun + current sun duration) meets the required drying time (timeNeedToDry, e.g., 240 seconds).

This ensures that even if sunlight is intermittent, total effective exposure time is tracked correctly.

2.4 Decision-Making Logic

Inside the main loop, the system performs the following evaluations:

2.4.1 Conditions to Bring Clothes Inside

The clothes are taken indoors if any of the following occur:

- Sunlight becomes insufficient (ldrReading ≤ LDRthreshold)
- Rain is detected (rainSensorReading ≤ rainSensorthreshold)
- Clothes have completed required drying time (isDried() == true)

If the clothes are currently drying outside (isSunDrying == 1), the system:

1. Stops time tracking
2. Sets isSunDrying = 0
3. Rotates the stepper motor anticlockwise to pull the clothes into the shelter

2.5 Motor Control (Clothes Movement)

Clothes movement is done using controlled multi-rotation steps:

2.5.1 Moving to Shelter (Anti-Clockwise)

The function rotateAnticlockwise() performs six full anti-clockwise rotations with short pauses, pulling the hanging rack indoors.

2.5.2 Sending Clothes Out to Dry (Clockwise)

The function rotateClockwise() performs six full clockwise rotations to move the clothes back outdoors into the sunlight.

Each rotation sequence includes:

- Status messages printed on Serial Monitor
- A delay between consecutive rotations for smooth operation

2.6 Continuous Monitoring

The loop executes every 500 ms and continuously:

1. Reads environmental sensor data
2. Checks drying completion status
3. Makes decisions based on current conditions
4. Moves the clothes automatically if needed

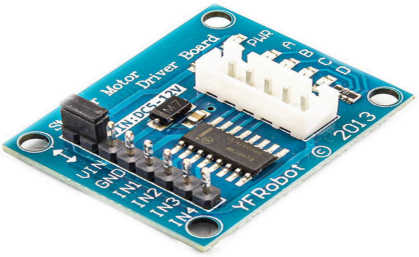
5. Updates time tracking accordingly

This ensures that the clothes are always protected from rain or low sunlight while maximizing drying efficiency.

3. Components



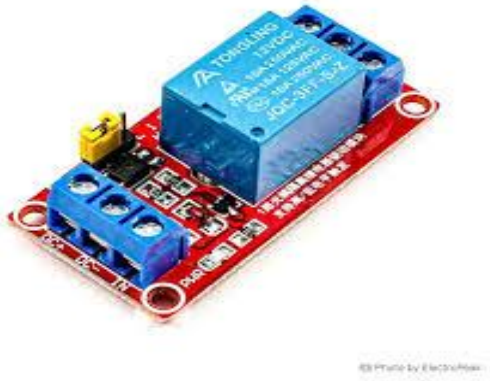
Arduino UNO



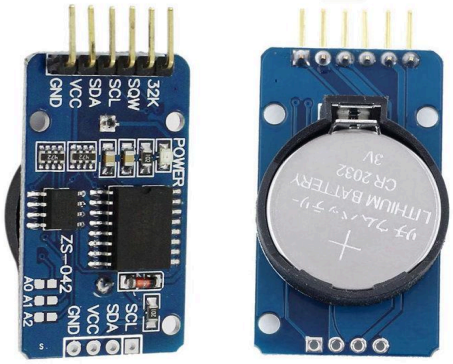
L298N Motor Driver



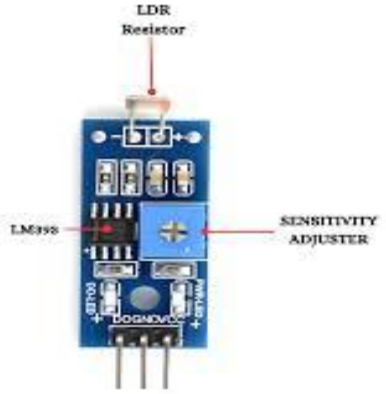
Rain Sensor Module



Relay Module



RTC Module



LDR

3.1 Rain Sensor

Working Principle

- The rain sensor detects water droplets using a resistive or capacitive sensing surface.
- When rain falls on the sensor board, water conducts electricity, reducing resistance between the traces.
- The sensor outputs an analog signal proportional to the amount of water — (low resistance = heavy rain).
- The signal is read by a microcontroller (like Arduino) to trigger actions (e.g., close window, stop irrigation).

3.2. Arduino Uno

Working Principal

- The Arduino Uno is a microcontroller board based on the ATmega328P chip.
- It works by reading input signals (from sensors, switches, etc.), processing them via programmed code, and sending output signals (to LEDs, motors, etc.).
- It follows the input → process → output principle.
- Power can come from USB or an external power source (7–12V).

3.3 LDR

Working Principal

- An LDR's resistance changes with light intensity.
- In bright light, resistance is low → more current flows.
- In darkness, resistance is high → less current flows.
- Arduino reads this change (as analog voltage) to detect light levels.

3.4 ULN2003 Motor Driver

Working Principal

- The ULN2003 is a Darlington transistor array used to drive stepper motors or relays.
- It acts as an interface between low-power Arduino pins and high-current loads (like motors).
- When Arduino sends a small control signal to an input pin, ULN2003 switches a higher current to the motor coil.
- It also contains flyback diodes to protect circuits from voltage spikes.

3.5 5V Stepper

Working Principal

- The 5V stepper motor works by converting electrical pulses into precise mechanical movement.
- When the Arduino sends a sequence of pulses through a driver (like ULN2003), the motor coils are energized in a specific order, creating magnetic fields that rotate the rotor step by step.
- Each pulse moves the motor a fixed angle, allowing accurate control of position, speed, and direction.
- It is used in this project to roll or unroll the cloth cover automatically during rain detection.

3.6 RTC

Working Principal

- The RTC uses a **quartz crystal oscillator** to generate a stable and precise clock signal for timekeeping.

- It maintains continuous tracking of 240 seconds (till the cloth dries) even when the main system is powered off.
- A small **backup battery** or power source ensures uninterrupted operation and preserves time data during power loss.

Drive Link:

(includes project demonstration video and pictures)
CI Project